

How well do you know the CERN & Society Foundation?

CERN community: here's your chance to win a limited-edition CERN & Society Foundation 10th anniversary isothermal bottle



The 2023 Non-Member State summer students. (Image: CERN)

Did you know that the [CERN & Society Foundation](#) is celebrating its 10th anniversary this month?

Created in June 2014 to support and promote CERN's mission and its benefits to the wider public, the CERN & Society Foundation works nationally and internationally across [three main areas](#): education and outreach, innovation and knowledge exchange, and culture and creativity.

- **Education and outreach** includes projects such as the [Beamline for Schools competition](#), the [CERN Non-Member State PhD Studentship Scheme](#) and the [Non-Member State Summer Students programme](#).
- **Innovation and knowledge exchange** encompasses several projects, including: [CERN-MEDICIS](#), which enables researchers to develop new approaches to fight cancer; TimePix, which is a detector for monitoring radiation in the environment and has an educational component for students; and Zenodo, an open source tool, built and developed at CERN to ensure that everyone can join in with open science.
- **Culture and creativity** projects include the [Arts at CERN programme](#).

A key achievement of the Foundation is of course the [CERN Science Gateway](#): as part of the Foundation's portfolio, the CERN Science Gateway would not have been possible without the generous support of its sponsors.

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A Word from Fabiola Gianotti

News from the June 2024 CERN Council Session

The European Strategy for Particle Physics, CERN's Medium-Term Plan and JINR were among many topics discussed in the 217th session of the CERN Council, find out more at the 2 July Directorate meeting

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A Word from Fabiola Gianotti

News from the June 2024 CERN Council Session

The European Strategy for Particle Physics, CERN's Medium-Term Plan and JINR were among many topics discussed in the 217th session of the CERN Council, find out more at the 2 July Directorate meeting

On 21 June, the CERN Council concluded a busy and fruitful June Session, during which several important discussions were held and decisions made.

Firstly, the delegates congratulated CERN and its personnel on the very good performance of the accelerator complex, the many beautiful physics results obtained by the LHC experiments and others, and the excellent progress made across the full spectrum of CERN's activities.

The Council approved CERN's Medium-Term Plan for the period 2025–2029, which describes the Organization's scientific and financial strategy for the coming years.

Preparing for CERN's future, the Council elected Professor Karl Jakobs, University of Freiburg, as Strategy Secretary for the next update of the [European Strategy for Particle Physics](#) and established both the Strategy Secretariat and the European Strategy Group. The overall timeline of the Strategy update was approved by the Council at its March session, setting a deadline of 31 March 2025 for the submission of community input and the goal of concluding the process in June 2026.

An important item on the June Council agenda was CERN's International Cooperation Agreement with the Joint Institute for Nuclear Research (JINR), an intergovernmental organisation located in Dubna,

Russia, with which CERN has collaborated since 1957. As the Council decided not to terminate the agreement, the participation of JINR in CERN's activities continues. However, the measures concerning JINR adopted by the Council in March 2022 remain in place.

The productive June Council Session again demonstrated the strong and sustained support of the Member and Associate Member States for CERN's mission and community and we are very grateful to them for this. We also express our heartfelt thanks to everyone throughout the CERN community for your hard work and dedication, which is at the core of what makes CERN so unique.

You are cordially invited to a meeting with the Directorate in the Science Gateway on Tuesday, 2 July from 11:00–12:30 where more information will be provided about the June Council Session and the European Strategy update. Please register by Friday, 28 June at this [Indico page](#) if you plan to attend the meeting and the refreshment that will take place afterwards. Those who are not present in person will have the possibility to submit questions during the meeting through a link that will be added to the Indico agenda just before the event.

Fabiola Gianotti
CERN Director-General

>>> *Continued from page 1*

"Thanks to our donors, we have been fostering education, technology and the arts for the last ten years. We have given talented students, teachers and artists from all over the world access to CERN's groundbreaking facilities and world-class education. The Foundation was created not only to awaken the youngest minds to the beauty of science but also to reinforce the existing dialogue

between CERN and society at large," said Pascale Goy, Head of the CERN & Society Foundation.

You can find out more about the Foundation via [its website](#) and its [latest Annual Review](#). To find out how you can support the Foundation, [click here](#).

How to enter the CERN & Society Foundation anniversary game:

- You must have a CERN email address to take part in this game.

- Watch the [10th anniversary impact video](#), and add together all of the impressive numbers that appear in the on-screen text.
- Send your total number to bulletin-editors@cern.ch by Sunday, 7 July at 11.59 p.m. CEST.
- Three winners will be drawn at random from among those who answer correctly. They will

each win a CERN & Society Foundation water bottle.

- The three winners will be announced in the next CERN Bulletin.

Good luck!

Anaïs Schaeffer

Accelerator Report: Technical stops always hold a few surprises

Restarting the entire accelerator complex after a technical stop is rarely seamless and often comes with unexpected technical challenges

The LHC technical stop started on 10 June, and the injector complex's technical stop two days later. These stops were scheduled to align with planned work on the Swiss electrical network. The planned work by [Swissgrid](#), the Swiss electricity transmission grid operator, could have caused fluctuations on the network, potentially disrupting or damaging the accelerator subsystems.

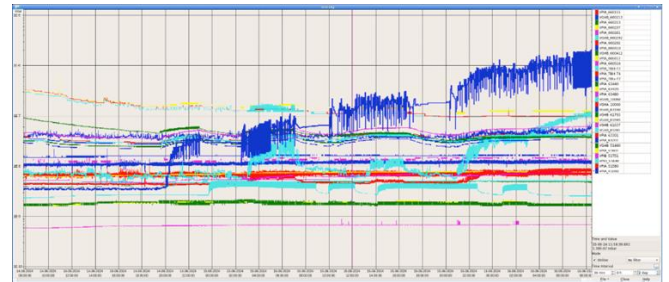
Swissgrid scheduled their work from Wednesday, 12 June to Friday, 14 June (Saturday being reserved as a back-up day in case additional time was needed). CERN's technical teams worked hard to complete their activities in the accelerator complex by 2.00 p.m. on the Friday, hoping to restart the complex early if Swissgrid finished ahead of schedule. At around 5.30 p.m., Swissgrid informed CERN that their work was complete. This notification marked the official restart of the accelerator complex.

Restarting the entire accelerator complex after a technical stop is rarely seamless. It can be particularly challenging late on a Friday afternoon... Nevertheless, with excellent support from standby services and experts, the restart went rather well. By Friday evening, beams were circulating in most of the accelerators in the complex, including the LHC.

Following each technical stop, the LHC requires a brief period for revalidation and intensity ramp-up. This time, the revalidation also included corrections to address [the collimation hierarchy issue](#).

Unfortunately, the process was interrupted on Sunday, 16 June, in the morning, when a vacuum leak was discovered in the SPS (see picture). This

required the replacement of a magnet, which was scheduled for the following day. Despite the leak, beams could still be provided to the LHC and the SPS North Area experiments during the night.



The vacuum pressure in several parts of the SPS. In royal blue, the vacuum pressure in a magnet with a leaking vacuum chamber. Two things can be observed: 1) the vacuum pressure is increasing over time, and 2) the ripple on the vacuum pressure signal is synchronous with the pulsing of the magnet. This means that, as the magnetic field of the magnet increases, the vacuum leak area opens wider. (Image: CERN)

On 17 June, technical teams replaced the magnet and, by 4.30 p.m., the magnet with the leaking vacuum chamber had been removed from the SPS tunnel. The newly installed spare magnet and vacuum chamber were connected, and vacuum pumping began. By 18 June, at around 10.00 a.m., the vacuum pressure was low enough to resume beam operation. However, a new issue, this time with the accelerating radiofrequency (RF) cavities, prevented beam acceleration in the SPS. After work by the RF experts, beam acceleration was possible by early evening, allowing the LHC revalidation to continue.

The LHC revalidation and intensity ramp-up were completed by the early hours of 20 June, allowing luminosity production to resume. However, when

the ATLAS magnets were ramped up to their nominal magnetic field, clogging was discovered in the main refrigerator cold box. This required a warm-up and purging cycle, which began that morning and finished on 25 June.

During the weekend, after completing the intensity ramp-up, luminosity production resumed. However, to prevent a significant difference in integrated luminosity between

ATLAS and CMS, it was agreed to bring forward some scheduled machine development (MD) activities and reduce luminosity production until the ATLAS magnet was fully operational again. These MD activities will be deducted from the next MD block, which is scheduled to start on 19 August, and will be replaced by luminosity production.

Rende Steerenberg

CERN70: A two-stage rocket

Part 12 of the [CERN70](#) feature series. Ted Wilson was involved in the design of the Super Proton Synchrotron (SPS) and played a leading role in its commissioning



The metal structure of the shuttering used when pouring the concrete walls of the SPS tunnel gave this remarkable optical effect. (Image: [CERN](#))

In the early 1960s, when the [Proton Synchrotron \(PS\)](#) had just come into service, the scientific community was already thinking about a machine ten times more powerful. But a new accelerator could be built elsewhere, with a new laboratory. The project soon found itself the object of a diplomatic and financial deadlock. [John Adams](#), the project leader, proposed a less expensive and more powerful alternative. Similar to a multi-stage rocket, he suggested using the PS as the injector for the new accelerator, to achieve higher energy at lower cost.

Thanks to this new impetus, the programme was approved in 1971. A few months later, work began to dig a 7-kilometre tunnel 40 metres underground, straddling the French-Swiss border, and to build a new laboratory in France on a 412-hectare site. While the construction work for the [Super Proton Synchrotron \(SPS\)](#) was under way, CERN was divided into two separate laboratories, each with its own administrative structure and its own [Director-General](#). The two were brought back together as a single laboratory in 1976. The accelerator ring consisted of more than a thousand magnets, which saw the commissioning of an ultra-modern control system for its time: 24 small control computers communicated via a high-speed transmission system between the tunnel and the control room. The computers were equipped with the [first capacitive touch screens](#), a technology that would be used in smartphones many years later.

On 17 June 1976, the day of the CERN Council meeting, the protons made their first turns around the ring at an energy of 300 GeV. Only a few hours later, the teams pushed the machine to 400 GeV, 100 GeV higher than the energy initially planned.

The programme of experiments got under way early the following year. The SPS supplied two experimental zones: the West Experimental Area on the Meyrin site and the new North Experimental Area on the Prévessin site. In 1981, the SPS was converted into a proton–antiproton collider, before becoming the second stage of an even larger “rocket” in 1989, the [Large Electron Proton collider \(LEP\)](#), 27 kilometres in circumference.

Recollections

Ted Wilson worked on the SPS design from 1967 and played a leading role in the commissioning of the Super Proton Synchrotron. He was involved in turning the SPS into a Storage Ring before leaving to join the running in of the Antiproton Accumulator in 1981. Subsequently he was head of the CERN Accelerator School for 12 years.



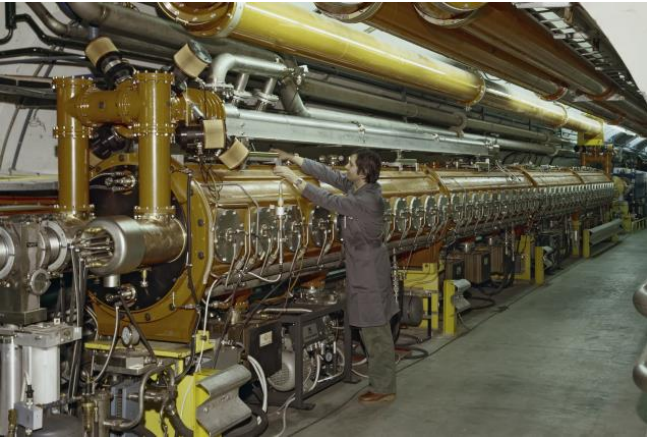
Ted Wilson takes a phonecall in the SPS control room in 1977. (Image: [CERN](#))

“As the Intersecting Storage Rings was nearing completion at the end of the 1960s, Europe’s next accelerator, a big synchrotron, was still a paper study. In the US, Fermilab’s new machine was

almost finished, built in half the time and for what seemed like half the budget. Several years had passed with no agreement on where to locate the new European accelerator and finally CERN Council could wait no longer. John Adams was invited back to get things moving.

I was a young CERN fellow, one of the tiny team of three working on the project, and I was more than surprised to be asked to redesign the entire machine to make it more compact and cheaper. Although we would show that we were well up with the latest technologies, Council would not be persuaded without detailed cost studies.

Then came John Adams’s masterstroke – the logjam over the site could be removed if there was a convincing reason for choosing one particular site. We were studying an option for a 400 GeV ring, which could fit at CERN, use existing machines as pre-accelerating injectors, and re-use some empty buildings as experimental areas. Building the machine at CERN would surely be cheaper than at any green-field site.



Inside the SPS tunnel in 1979. (Image: [CERN](#))

This “Project B” was a secret known only to one or two of us while John Adams toured Europe and, with painstaking diplomacy, disarmed national ambitions to host the new accelerator. My contribution to maintaining secrecy was to break a leg skiing and to spend six weeks under traction far from a phone or computer. When the time was ripe, a rather impatient John Adams came to my bedside to tell me to get well soon. It was time to

remove the wraps on the new machine, which was to become known as the SPS.

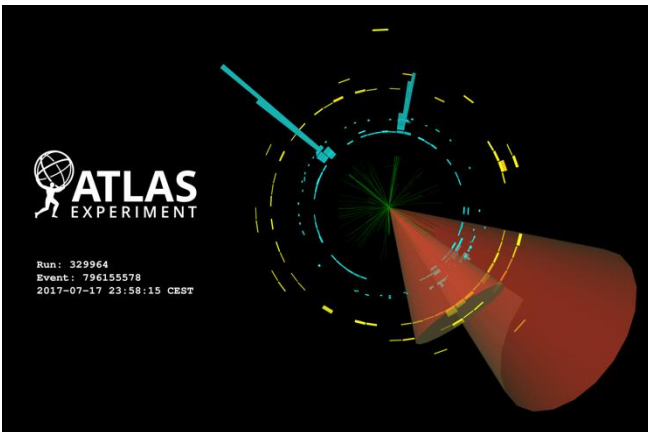
A pivotal moment for John Adams, who was from the United Kingdom, was his encounter with a certain Margaret Thatcher, visiting CERN as Minister of Science and Education. I have rarely seen John as excited as when he rushed along the corridor with the news: “Margaret has just phoned to tell me that the Cabinet has agreed the UK should rejoin the Project. Isn’t that nice of her?”

In 1976 the SPS came smoothly on-line as planned. Although five years behind its American rival, it was soon to be turned into a Nobel Prize winning facility, colliding protons against antiprotons, thanks to the brilliance of Carlo Rubbia and Simon van der Meer.”

This interview is adapted from the 2004 book “Infinitely CERN”, published to celebrate CERN’s 50th anniversary. Ted Wilson [died in 2016](#) at the age of 78.

ATLAS dives deeper into di-Higgs

By combining multiple Higgs boson pair studies, physicists are closer to finding out how the particle interacts with itself, providing clues to the stability of the Universe



An event display of a di-Higgs candidate event taken in 2017. (Image: ATLAS collaboration/CERN)

Remember how difficult it was to find one Higgs boson? Try finding two at the same place at the same time. Known as di-Higgs production, this fascinating process can tell scientists about the Higgs boson self-interaction. By studying it, physicists can measure the strength of the Higgs boson’s “self-coupling”, which is a fundamental

aspect of the Standard Model that connects the Higgs mechanism and the stability of our Universe. Searching for di-Higgs production is an especially challenging task. It’s a very rare process, about 1000 times rarer than the production of a single Higgs boson. During the entire Run 2 of the Large Hadron Collider (LHC), only a few thousand di-Higgs events are expected to have been produced in ATLAS, compared with the 40 million collisions that happened every second. So how can physicists find these rare needles in the data haystack? One way to make it easier to look for di-Higgs production is to search for it in multiple places. By looking at the different ways di-Higgs can decay (decay modes) and putting them together, physicists are able to maximise their chances of finding and studying di-Higgs production.

Researchers at the ATLAS collaboration have now released the most sensitive search for di-Higgs production and self-coupling yet, achieved by

combining five di-Higgs studies of LHC Run 2 data. This new result is their most comprehensive search so far, covering over half of all possible di-Higgs events in ATLAS.

The five individual studies in this combination each focused on different decay modes, each of which has its pros and cons. For example, the most probable di-Higgs decay mode is into four bottom quarks. However, Standard Model QCD processes are also likely to create four bottom quarks, making it difficult to differentiate a di-Higgs event from this background process. The di-Higgs decay to two bottom quarks and two tau leptons has moderate background contamination but is five times less common and has neutrinos that escape undetected, complicating physicists' ability to reconstruct the decay. The decay to multiple leptons, while not too rare, has complex signatures. Other di-Higgs decays are even more rare, such as the decay to two bottom quarks and two photons. This final state accounts for only 0.3% of total di-Higgs decays but has a cleaner signature and much smaller background contamination.

By combining the results from searches for each of these decays, the researchers were able to find

that the probability that two Higgs bosons are produced excludes values more than 2.9 times the Standard-Model prediction. This result is at 95% confidence level, with an expected sensitivity of 2.4 (assuming that this process is not present in nature). They were also able to provide constraints on the strength of the Higgs boson self-coupling, achieving the best-yet sensitivity on this important observable. They found that the magnitude of the Higgs self-coupling constant and the interaction strength of two Higgs bosons and two vector bosons are consistent with Standard Model predictions.

This combined result sets a milestone in the study of di-Higgs production. Now, ATLAS researchers have set their sights on data from the ongoing LHC Run 3 and upcoming High-Luminosity LHC operation. With this data, physicists may be able to observe the elusive Higgs-boson-pair production at last.

Read more:

[ATLAS Briefing Paper](#)

ATLAS collaboration

CERN's communications receive European recognition

CERN's Education, Communication and Outreach group receive the European Association of Communication Directors Excellence award



Ana Godinho, head of CERN's Education, Communication and Outreach group receiving the award at the EACD Annual Summit surrounded by EACD board members Andrew Hillier and Susan Kelly. (Image: EACD)

The European Association of Communication Directors (EACD) has awarded CERN's Education, Communication and Outreach group the EACD 2024 Communications Excellence Award.

The award is given each year to individuals or groups who demonstrate excellence and best practice in communications and who build bridges through effective international communications, contributing to broader society.

The jury rewarded CERN's communication in particular for its commitment to transparency, knowledge sharing and transnational communications.

"In its 70th birthday year CERN continues to lead not just in scientific research, but in engaging the

world and communicating its discoveries and innovation”, said Kim Larsen, the EACD President. “This award is well-deserved recognition of the work done by everyone involved in CERN communications and the many scientists we work

with every day”, explains Ana Godinho, head of the Education, Communication and Outreach group, who received the award at the EACD Annual Summit on 27 and 28 May in Brussels.

Students from Estonia, Japan and the USA win the 11th edition of Beamline for Schools

Three teams of secondary school pupils from Estonia, Japan and the United States have been selected to carry out their own experiments using accelerator beams at CERN and DESY



Winners of the 2024 CERN Beamline for Schools competition: *Sakura Particles*” from Japan (left), “*Mavericks*” from Estonia (top right) and “*SPEEDers*” from the USA (bottom right). (Images: *Sakura Particles*, *Mavericks*, *SPEEDers*)

Geneva and Hamburg, 25 June 2024. [Beamline for Schools \(BL4S\)](#) is a physics competition run by [CERN](#), the European laboratory for particle physics, open to secondary school pupils from all around the world. Participants are invited to prepare a proposal for a physics experiment that can be undertaken at the beamline of a particle accelerator, either at CERN or at [DESY](#) (Deutsches Elektronen-Synchrotron in Hamburg, Germany). In 2024, three winning teams have been chosen, based on the scientific merit of their proposal and the communication merit of their video.

“*Mavericks*”, a team from the [Secondary School of Sciences](#) in Tallinn and the [Hugo Treffner Gymnasium](#) in Tartu, Estonia, and the team “*Sakura Particles*”, which brings together pupils from [Kawawa Senior High School](#) in Kanagawa, [Joshigakuin Senior High School](#) and [Junten High](#)

[School](#) in Tokyo, [Kawagoe Girls High School](#) in Saitama and [Kitano High School](#) in Osaka, Japan, will travel to CERN in September 2024 to perform the experiments that they proposed. The team “*SPEEDers*” from [Andover High School](#) in Andover, USA, will carry out their experiment at a DESY beamline.

A beamline is a facility that provides high-energy fluxes of subatomic particles that can be used to conduct experiments in different fields, including fundamental physics, material science and medicine.

BL4S started in 2014 in the context of CERN’s 60th anniversary. Over the past 10 years, more than 20 000 pupils from all over the world have taken part in the competition, and 25 teams have been selected as winners. The participation rate has been rising consistently over the years, with a record 461 teams from 78 countries submitting an experiment proposal in 2024.

“Preparing a proposal for a particle physics experiment is a very challenging task. The success of Beamline for Schools shows that, when provided with the right support, high-school students can design feasible, interesting and imaginative experiments,” says Charlotte Warakaulle, CERN Director for International Relations. “We are continuously impressed by the quality of the proposals, and this year is no exception. The candidates demonstrated impressive creativity and great rigour, two essential qualities for students who might decide to take up scientific careers.”

The fruitful collaboration between CERN and DESY started in 2019 during a long shutdown period of the CERN accelerators. This is the sixth

year that the German laboratory has hosted competition winners.

“Every year I am very impressed by the creativity and determination of the team members,” says Beate Heinemann, Director in charge of Particle Physics at DESY. “I am already looking forward to hosting the team from the USA this year. This programme is so important to me as it advances not only science but also the cultural exchange between young people from different nations.”

“Our experiment will focus on detector development for high-altitude ballooning applications,” says Saskia Pöldmaa, one of the “Mavericks” members, from Estonia. “This is by far the biggest opportunity we have had so far in our lifetime so we will hold onto it dearly. We can’t wait to calibrate our homemade muon detector!”

“Our team focuses on detector development for muon tomography applications. We will test and optimise our homemade two-dimensional position-sensitive detector,” says Chiori Matsushita from the Japanese “Sakura Particles” team. “CERN has always been a dream for us. Finally getting to go there, not as a tourist but to do experiments, is amazing!”

“We focus on beam diagnostics: our aim is to measure and analyse the Smith-Purcell (SP) radiation emitted by different diffraction gratings when DESY’s electron or positron beams pass by,” says Niranjana Nair from the US “SPEEDers” team. “We are thrilled to have the opportunity to not just watch scientific advancement passively, but actively contribute to it at DESY: the ultimate goal of our experiment is to research SP radiation as a tool for beam diagnostics.”

The winning proposals were selected by a committee of CERN and DESY scientists from a shortlist of 49 particularly promising experiments. In addition, three teams will be recognised for the most creative video proposals and another 13 teams for the quality of physics outreach activities they are organising in their local communities, taking advantage of the knowledge gained by participating in BL4S.

Beamline for Schools is an education and outreach project funded by the [CERN & Society Foundation](#)’s donors. This 11th edition is supported notably by ROLEX through its Perpetual Planet Initiative and by the Wilhelm and Else Heraeus Foundation.

New beam dumps: cut along the dotted line

Dismantling a beam dump is a real conundrum – that’s what the 2020 “autopsy” concluded. To solve it, the new spare LHC beam dumps have been made easier to “cut”

When particle beams circulating in the LHC need to be stopped, they are directed towards the beam dumps. In 2020, during Long Shutdown 2, the main LHC beam dumps, which had been in place in the accelerator from the start, [were replaced](#) by spare dumps – themselves heavily modified with respect to the original design – because they were showing signs of wear and tear. To withstand the onslaught of the current run, Run 3, these spares had been upgraded before being installed as the main beam dumps. They are still in place today and operating successfully.

The [autopsies](#) carried out on the first beam dumps highlighted their strengths and weaknesses. Those findings, coupled with [additional studies](#) carried out at HiRadMat, led to a strategy for designing

new spare beam dumps and the future HL-LHC beam dumps.

A tried and tested recipe

Originally, the beam dumps were made of three main materials: blocks of high-density graphite and discs of low-density graphite held in place by two discs of extruded graphite. “The autopsies revealed that both of the extruded graphite discs were cracked, while the high- and low-intensity graphites were generally in good condition, thereby validating their use in the beam dumps for Run 3,” recalls Nicola Solieri, project engineer in charge in the SY department’s Targets, Collimators and Dumps section ([SY-STI-TCD](#)).

The new spare beam dumps therefore comprise these two components: six blocks of high-intensity

graphite and 1700 discs of low-density graphite, now contained between two carbon-fibre-reinforced carbon discs – a material [tested and validated at HiRadMat](#) – which replace the old extruded graphite discs.



Image 1: Each high-density graphite block is heat-shrunk inside a steel cylinder. The cylinder is heated to 300 °C to expand it, then slipped over the graphite block. The cylinder contracts as it cools, tightening around the graphite block like a skin. This technique was carried out at CERN for the first time on an object of this size. (Image: CERN)

A novel design: easier to “cut”

“The autopsy and disposal of the original beam dumps brought to light a major challenge: how to dismantle them,” says Marco Calviani, SY-STI-TCO section leader. “In 2021, cutting them open was a really [tricky operation](#), in particular because of their radiation levels. In the end we managed to find a solution using CERN’s expertise, but it was clear that we would need to address this issue for future beam dumps.” To do so, the team in charge of the project came up with a “detachable” beam dump design (see images 1–3). “To make it easier to dismantle the beam dumps in the future, as well as making the LHC’s operation more sustainable and improving waste management, we wanted to keep the number of welds to a minimum, and we left a 5-cm space between each of the components to make it easier to take them apart,” explains Nicola Solieri. These measures will greatly facilitate dismantling operations in the event that the spare beam dumps do end up being used and thus become radioactive. And if they are not used in the LHC, their components will be easier to reuse for the HL-LHC beam dumps.

“The whole process of developing these new beam dumps, from design to assembly, was conducted in house with CERN’s competences, which was not the case with the original beam dumps,” underlines Damien Grenier, technical engineer in charge of the assembly of the dumps. “This way, we ensure that CERN has all the know-how and skills required for the key stages of beam dump production and assembly, ready for the HL-LHC.” Indeed, challenges abound for the design of the HL-LHC beam dumps, which will be subject to unprecedented beam intensities. A complex R&D programme is under way with a view to identifying and validating new materials for the beam dump core and container, as well as new assembly techniques.

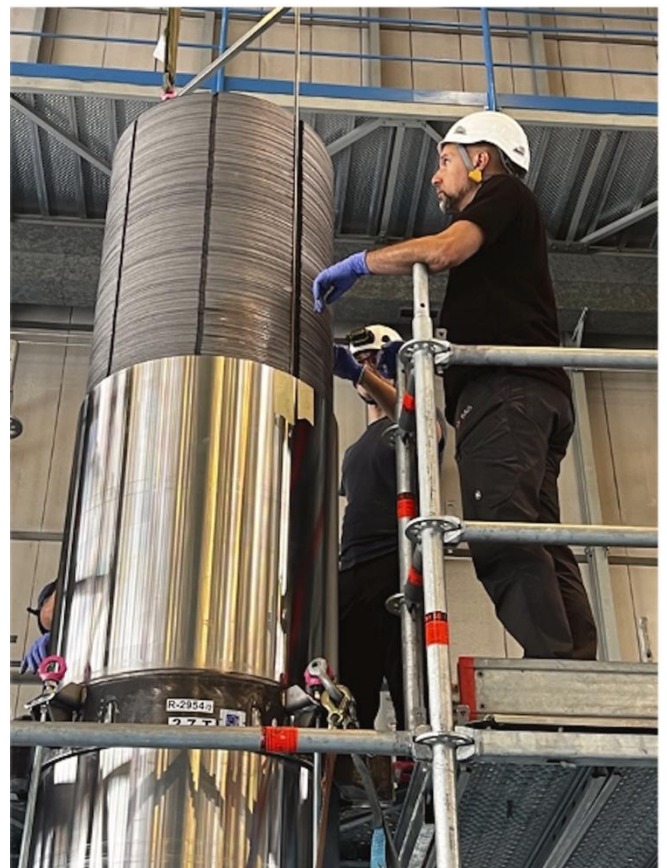


Image 2: With the beam dump tube in a vertical position, the teams from the SY-STI-TCO section insert the 1700 low-density graphite discs, which are held in place at either end by rings, instead of being welded. (Image: CERN)



Image 3: The 13 “slices” of the beam dump are welded together end to end in CERN’s main workshop, an operation coordinated by the Engineering department’s Mechanical and Materials Engineering group (EN-MME). One of the

carbon-fibre-reinforced carbon discs can be seen at the end of the tube. (Image: CERN)

Meanwhile, the spare beam dumps were finalised in autumn 2023 and installed in their storage cavern at the beginning of 2024. “The two spare beam dumps were produced in record time thanks to the involvement and the tireless efforts of many colleagues from various CERN groups and departments,” continues Marco Calviani. “This is further proof that many challenges can be overcome in house thanks to the team spirit that reigns at CERN and using materials developed by industry.”

Anaïs Schaeffer

Work Well Feel Well: Every successful person has allies

Part five of the [Work Well Feel Well](#) series looks at the benefits of support networks



Seeking support

It can be easy to dwell on negative interactions, for example, if someone criticises or annoys us. We end up absorbing other people’s stress, and negativity becomes far more present in our lives than positivity.

Instead of taking on this negative energy, why not reframe it and make a conscious choice about whom or what we let into our lives. This means choosing allies who can support us intellectually, emotionally or practically.

An ally can be defined as a person, colleague, friend or guide who inspires because...

- **Allies offer intellectual support:** Allies can offer expertise and experience to help find a simple solution to a problem. For example, more experienced colleagues can guide newer team members in complicated situations, showing

them shortcuts, helping them to avoid traps and warning them of potential dangers.

- **Allies offer emotional support:** Our emotions can be heightened by a difficult situation, leading to feelings of fear, excitement, anxiety, confusion, despondency or discouragement. A confidential debrief with a close colleague, for example, can offer us a different perspective; they see the situation from a healthy distance and can encourage us to think things through.
- **Allies offer practical support:** Allies can offer discreet, irreplaceable support, without which we would not be able to complete a task. This could be, for example, a supervisor supporting a team by reframing roles, responsibilities or complex tasks.
- **Allies can also inspire us:** Allies can be role models who, with their values and qualities, guide us in our personal development. We can no doubt all think of an inspirational teacher from our childhood who was a hero not for who they were but for what they inspired in us.

When we stop to reflect on our network of allies, we can appreciate how indispensable it is to our efficiency at work. Alliances are formed, strengthened and developed if we nurture them; we can then act with inspiration, with assistance and with support. We can also be an ally ourselves, checking in on colleagues and friends and offering

a space in which they can give a more meaningful response than “I’m fine”.

Take action

As part of the “Efficiency and caring at work” campaign, the [Work Well Feel Well website](#) now offers **useful resources** that can be downloaded, including an exercise on **mapping** our professional and private **network of allies**.

You can build your network of allies via:

- **Informal networks:** The Diversity and Inclusion website lists several [informal networks](#) relating to nationality, gender and more, including the new [Supporting Neurodiversity at CERN \(SNAC\)](#)

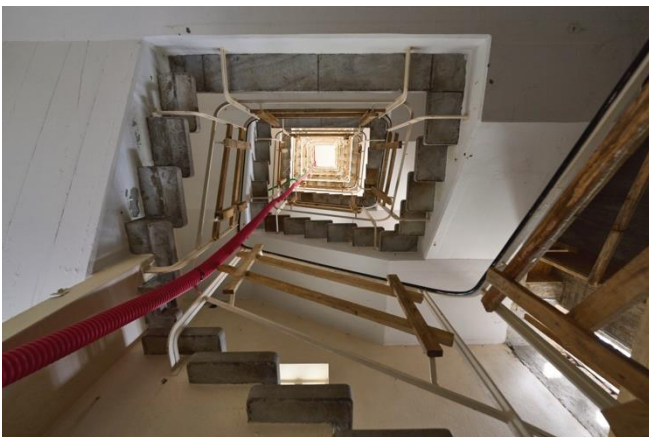
network. The Women in Technology network also runs a [mentoring scheme](#).

- **CERN clubs:** The CERN Staff Association website lists more than 40 different [CERN clubs](#).
- **Training courses:** The [CERN training catalogue](#) offers language, leadership, safety, technical and personal development training courses. Not only do you learn something new, but you can also meet potential allies from across the Organization. In particular, the [Onboarding – Connecting the Dots](#) sessions can help you to develop your network within CERN.
- **Ethics at CERN:** The [Ethics at CERN](#) webpage includes a [list of options for help and support](#)

HR department

Building 60 renovations: one year on

As the Main Building enters the next phase of its renovation, find out what’s changed and why



CERN’s iconic Main Building, Building 60, has reached a crucial milestone in its two-and-a-half-year renovation, with the completion of the so-called remediation phase. The overall objectives have been to bring the building up to current regulatory standards following the strictest possible safety measures and to improve energy efficiency to the highest environmental standards, while preserving the building’s exceptional architectural value.

Since summer 2023, Building 60 has been surrounded by scaffolding and confined for this remediation phase, involving the removal of mainly asbestos as well as several other pollutants. Planning had already started in 2021, with a specialist consultant assisting the SCE and

HSE teams with analyses. Once Building 42 was prepared for the Management and other displaced teams, Building 60 was fully emptied and set up, with only structural elements kept as a foundation for the renovation.

A diagnosis of pollutants helped determine more precisely the type, location and quantities for removal, while the creation of a test site made it possible to better understand the design of the building and the distribution of pollutants, in particular flocked asbestos. The pollutant removal works also included disposal of the waste generated. Ethically, CERN’s priority was to ensure that the flocked asbestos waste would not go to landfill. A novel technology, called “inerting” (“inertage” or “vitrification” in French), was used instead, which involves incinerating the waste by plasma torching in a dedicated facility in France: the process transforms it into a glass-type material, which can be reused in the form of aggregates as a road underlay.

Teams faced a range of challenges due to the building’s layout and accessibility, parallel works and activities, waste removal and inevitable timescale management issues, as well as heatwaves and heavy rain. Given that this project impacts so many people, the CERN community’s patience throughout has been highly appreciated.

Mar Capeans, SCE Department Head, commended the efforts so far, saying: “It takes a brave team to make it possible: the Management, who agreed to spend half their mandate far from the tower, HSE and SCE who worked together closely from the preparation phase and throughout the execution, and the many teams and services, inside and outside CERN, who have made achieving this milestone with such excellent results possible.”

The next stage of the [renovation and rehabilitation of the building](#) now begins, so that by mid-2025 the Main Building will be restored to its former glory, freshened up, safe and compliant in all respects.

SCE department & HSE unit

Arts at CERN awarded European Commission’s S+T+ARTS Grand Prize for Innovative Collaboration

Arts at CERN wins the Grand Prize for groundbreaking initiatives at the nexus of science, technology and the arts



Swiss artists Andrea Anner and Thibault Brevet (AATB) with scientist Laura Lambert at the MEDICIS facility during their residency with Arts at CERN. (Image: CERN)

The European Commission’s [S+T+ARTS initiative](#) has awarded [Arts at CERN](#), the arts programme of the Laboratory, the prestigious Grand Prize for Innovative Collaboration in recognition of its efforts to establish transformative collaborations that connect society with science and research.

Two S+T+ARTS Prizes are awarded annually – one for Innovative Collaboration and one for Artistic Exploration – to outstanding projects that show significant impact on social and economic innovation at the convergence of science, technology and the arts, with each receiving €20 000 in prize money. The Grand Prize ceremony will take place on 5th September at the 2024 Ars Electronica Festival in Linz, Austria, where Arts at CERN will present an exhibition highlighting the collaborative nature of its programmes. As CERN

commemorates its 70th anniversary, the exhibition will showcase the Laboratory as a unique environment for creativity, collaboration and artistic inquiry.

The mission of Arts at CERN – a programme funded by the [CERN & Society Foundation](#) – is to facilitate dialogue between artists and the CERN community. Since the launch of the first artist residency in 2012, over 250 artists from around the globe have been invited to experience how fundamental science can address the unresolved questions about our universe, engaging with over 1000 scientists and the entire vibrant community of CERN.

Initially focused on artist residencies, Arts at CERN’s programmes have evolved to encompass art commissions, exhibitions and events that bridge the gap between art, science and society. Guided by the vision that science forms an integral part of contemporary culture, Arts at CERN has cultivated a global collaboration network, partnering with scientific laboratories, cultural organisations and research centres.

“Art and science are essential pillars of society, and artists and scientists share common core-values: curiosity and the passion to understand the world on a deeper level. At CERN, we have always believed that these two fields could grow together through creative interaction, and that is the basis of Arts at CERN. The S+T+ARTS Prize acknowledges the effort of the team to create meaningful encounters between artists and

scientists and recognises Arts at CERN as a leading programme for engagement between art and science”, says Charlotte Warakaulle, Director for International Relations at CERN.

Since 2018, Arts at CERN has supported 35 art commissions, enabling artists to embark on creative explorations alongside scientific partners and experiments. In return, scientists gain fresh perspectives on their research through the transformative lens of artistic expression. A notable milestone was achieved with the recent launch of the CERN Science Gateway, which features new commissions by former artists in residence at CERN that seek to inspire both the scientific community and the public.

Through Arts at CERN, the Laboratory continues to explore the profound significance of fundamental research in our society. By offering spaces for artistic inquiry within the research context, CERN sets itself apart as a unique place for cultural innovation.

About S+T+ARTS

S+T+ARTS is a large-scale initiative of the European Commission that aims to bring technology and artistic practice together in a way that benefits

both European innovation policy and the art world. The initiative promotes and supports people and projects that contribute to tackling Europe's social, environmental and economic challenges.

S+T+ARTS Prize

Part of the S+T+ARTS initiative is the prestigious S+T+ARTS Prize, which is awarded to two projects annually, each of them receiving €20 000 in prize money. The competition honours innovative projects at the nexus of art, technology and science that contribute to economic and social innovation.

The S+T+ARTS Prize is coordinated by Ars Electronica, on behalf of the European Commission, together with the consortium partners INOVA+, French Tech Grande Provence, Media Solution Center Baden-Württemberg, Salzburg Festival, Sonar, T6 Ecosystems and Kustodie at TU Dresden University of Technology. The jury for the 2024 S+T+ARTS Prize was composed of Francesca Bria, Fumi Hirota, Manuela Naveau, Katja Schechtner and Miha Turšič.

Instruments of Vision opens in Santiago de Compostela as a collaboration between Arts at CERN and IGFAE

In this exhibition, Armin Linke, former artist in residence at CERN, explores the research environments at the laboratories

The exhibition [*Instruments of Vision*](#) comprises photographs and videos taken by Armin Linke during visits to experimental facilities, such as CERN or the Laboratori Nazionali del Gran Sasso (L'Aquila, Italy), or the Institute for Quantum Optics and Quantum Information (Vienna, Austria), since 2000.

For this occasion, Armin Linke has produced images that portray some of the work pursued by staff from the Galician Institute of High Energy Physics (IGFAE) at CERN. The photographs show how the scientific community has generated very complex instruments that allow us to observe and understand how the most fundamental elements of matter work. These photographs are enriched by the unique location of the exhibition, which is

on display at Igrexa da Universidade, a baroque church at the heart of the old town.

In the exhibition, Linke invites visitors to witness spaces of research, where various kinds of instruments or components – such as parts of particle detectors, data processors archival images – can be found. His work captures the dynamic nature of laboratories, highlighting often overlooked elements and scenes in which physicists become intertwined with the precise instruments that underpin scientific inquiry. These activities are depicted not merely as isolated scientific endeavours but as integral components of a broader social and cultural composition, reflecting the interconnectedness of science, society and technology.

CERN science and history are explored in three interviews with key voices in the community: Maria Fidecaro, an experimental physicist and one of the first female scientists at CERN, Rolf-Dieter Heuer, former Director-General of CERN, and Peter Jenni, one of the founding fathers of ATLAS and a former spokesperson of the experiment. The three physicists shared with the artist their views on the development of detector and imaging technologies at CERN and the role these technologies play in advancing particle physics.

“Breakthroughs in physics over the past decades, guided by complex instrumentation and sophisticated experiments, have transformed our understanding of fundamental concepts such as

matter, space and time. As artists engage with science, our understanding deepens and becomes more diverse, inviting everyone to participate in an intellectual and creative exchange that takes place across disciplines”, writes Mónica Bello, Head of Arts at CERN and curator of the exhibition. Instruments of Vision opens on 21 June and runs until 28 August 2024. The exhibition commemorates the 25th anniversary of IGFAE and the 70th anniversary of CERN. On this occasion, IGFAE and Arts at CERN join forces to promote new models of dialogue between artists and scientists at the laboratories.

Ana Prendes

Computer Security: Dear summer students, welcome!

A warm welcome to the summer-student class of 2024! In order to make your digital life as comfortable as possible, here are a few things you need to know

A warm welcome to the summer-student class of 2024! We're glad that you made it to CERN! We offer a packed agenda for the next two months: challenging lectures; interesting projects to tackle with your team; and lots of time to take a great big gulp of CERN's academic freedom, spirit and creativity! In order to make your digital life as comfortable as possible, however, there are a few things you need to know.

When you join CERN, you're given a CERN computing account. Take care of [your account password](#) as any evil-doer might misuse it to spam the world on your behalf, abuse CERN's computing clusters in your name, download journals in bulk from CERN's digital library, or simply compromise your CERN PC and extract your photos, documents or personal data, or spy on you using your computer's microphone or webcam. Worst-case scenario, [the whole Organization is at risk!](#) Similarly, take good care of your CERN and personal computers, tablets and smartphones. Give them some freedom to update themselves so you benefit from [the latest protective measures](#). “Auto-update” is a good friend, just make sure that it's enabled – as it should be by default.

A particularly nasty way to lose your password, at CERN or at home, is to reply to so-called “phishing emails”, i.e. emails asking for your password. No serious person – the CERN Computer Security

team, the CERN Service Desk or your CERN supervisor – would send such an email, only dishonest people or fraudsters would. So stay on the lookout and don't enter your password in weird webpages. Don't click on links in emails [obviously not intended for you](#), for example, emails not addressed to you, [not coming from the real CERN](#), not written in one of your native languages, or of no relevance to you. Ask us at Computer.Security@cern.ch if you have any doubts. Similarly, don't randomly click on web links, but stop and think first. Otherwise, [you might infect your computer](#) in no time – and the sole remedy will be a full reinstallation of your device (easier if you have backups!).

CERN has awesome network connectivity to the world. But it's for professional purposes. While private usage is tolerated, please do not abuse this. Keep your bandwidth low. In particular, refrain from bulk downloading movies or software. Remember “copyright”? It also applies at CERN. [Any violation of copyright](#) reported to CERN will be followed up and any infringement costs will be passed on to the perpetrator. The same holds true for pirated software. If you have stored pirated licence keys on your device, it's time to delete them. Companies are monitoring for abuse of their software and infringement costs [can quickly reach five to six figures](#). This one

is of particular importance: if you need particular software, have a look at [CERN's central software repositories](#).

While you're at CERN, you might be working on a project requiring digital resources – setting up a webpage, writing some code, developing hardware. Please don't reinvent the wheel if you need a database. Or a webserver. Or some software. The CERN IT department can provide a wide variety of [centrally managed and secure services](#) for your digital convenience. Just put yourself on their shoulders and build on top. Free up your time and brain for creativity and let CERN IT provide the tools. Moreover, make sure that all your development work, software, design drawings, documentation and so on are made available to your supervisor when you leave. This will ensure your legacy lives on at CERN. If you keep them to yourself, they'll get purged and deleted, and your time at CERN will be forgotten.

Finally, like anywhere else, there are some rules to respect. Use of CERN's computing facilities is governed by the [CERN Computing Rules](#). Basically, be reasonable. Don't do anything that could be considered immoral, illegal or abusive. Similarly, personal use of CERN's computing facilities is tolerated, but within the aforementioned limits. For example, browsing pornography is forbidden unless you have a good professional reason to do so (and it might be awkward receiving a corresponding cease-and-desist email from us). In another example, [crypto-mining on CERN's computing clusters](#) is definitely a no-no. Just don't. So, make sure that you respect these few ground rules – keep your system up to date – protect your password – STOP-THINK-DON'T CLICK – respect copyright – preserve your work – follow the CERN Computing Rules. We wish you a great and exciting stay at CERN. Have fun and enjoy!

Computer Security Team

Official News

Members of the personnel shall be deemed to have taken note of official news published for the CERN community. Reproduction of all or part of this information by persons or institutions external to the Organization requires the prior approval of the CERN Management. The full list of official news is available here: <https://cern.ch/official-news>.

Staff rules and regulations - 11th edition

Modification no. 24

In accordance with recommendations made by the Finance Committee and decisions taken by Council in June 2024 (CERN/FC/6819-CERN/3827), please find below the page of the Staff Rules and Regulations which have been updated further to the modifications coming into force on 1 July 2024:
Chapter ii – conditions of employment and association

Section 1 – Employment and association
(modification of page 12)

The complete updated electronic version of the Staff Rules and Regulations is accessible via [CDS](#).

HR department

Announcements

Upcoming events

Check out this curated list of upcoming events for the CERN community

27 May–27 June | At CERN | Meyrin and Prévessin | CERN Table Tennis Club | EN & FR

[CERN 70th Anniversary Table Tennis Tournament](#)

14 Jun–14 Jul | At CERN | R1 | Novae | FR
[Euro 2024 matches on the big screen](#)

2 Jul 11:00 | At CERN | CERN Science Gateway | Directorate | EN & FR

[Directorate Meeting - July 2024 / Réunion d'information du Directoire - Juillet 2024](#)

2 Jul 14:00 | At CERN | Online | SCE | EN & FR

[6th CERN Service Management Forum / 6è Forum "CERN Service Management"](#)

2 Jul–2 Aug | Knowledge Sharing | Main Auditorium | Summer Student Programme | EN

[Summer student lecture programme 2024](#)

2–31 Jul | Knowledge Sharing | IT Amphitheatre | CERN openlab | EN

[CERN openlab summer student lecture programme 2024](#)

2–12 Jul | Knowledge Sharing | Palais des Nations | UN Geneva, CERN & ITU | EN

[Exhibition: Not a Woman's Job? Changing Faces of Science and Tech](#)

4 Jul | 20:00 | Knowledge Sharing | CERN Science Gateway | CERN70 | EN

[CERN70 public event: Exploring farther – Machines for new knowledge](#) (more [CERN70](#) events)

6 Jul 15:00 | At CERN | R3 terrace, Prévessin | Music Club | EN & FR

[Hardronic Music Festival 2024](#)

6–7 Jul | Knowledge Sharing | Parc de la Perle du Lac | Education, Communications and Outreach | FR

[Le CERN à la Nuit de la Science](#)

8–19 Jul | Physics | CERN | Theory Institute | EN

[Lattice@CERN 2024](#)

3 Aug 14:00 | At CERN | R1 Terrace, Meyrin site | Music Club | EN & FR

[Music on the Lawn](#)

Registration now open

22 Sep–5 Oct | Accelerators | Santa Susanna (ES) | CERN Accelerator School | EN

[Introduction to Accelerator Physics](#)

11 Oct 14:00 | Knowledge Sharing | Online | HR Learning and Development | EN

[L&D Micro-talk 12 – Demystifying creativity](#)

27 Nov–6 Dec | Experiments | CERN | DRD1 | EN

[DRD1 Gaseous Detectors School](#)

This is a curated list of events relevant to the CERN community. More events are available here: home.cern/events

Indico also shows ALL events happening [today](#), [this week](#) and in a [calendar view](#).

If you would like your event to appear on an upcoming Bulletin events list, please contact bulletin-editors@cern.ch

Sunshine: know the risks

Understanding the impact of the sun on your health and what you can do to protect yourself from its harmful effects



We all enjoy the benefits of sunshine in the summer, with numerous positive effects on our physical and mental health and well-being. It is no secret, though, that exposure to the sun carries a number of risks, from the effects of the heat, to developing skin cancer and other skin disorders, to harming your eyesight. Between 50 and 70 percent of skin cancer cases result from excessive exposure to sunlight and associated [ultra-violet radiation](#), which is neither seen nor felt.

Skin cancer is the 17th most common cancer worldwide, and Switzerland ranks among the top ten countries in terms of incidence rates ([WCRE, 2020](#)). According to the [WHO Global Cancer Observatory](#) (2022), the [incidence rate of skin melanoma](#) in Switzerland stands at 13.9 per

100 000, while the country's [incidence rate of non-melanoma skin cancer](#) is particularly high, with 37.5 per 100 000, compared with its European neighbours, which average some 10–20 per 100 000.

The risks posed by natural and artificial ultra-violet light are not necessarily known, and nor are effective prevention measures. Simple tips include limiting time in the midday sun, seeking shade, wearing appropriate clothing, a sunhat and sunglasses, and applying high-factor sun cream. These all provide a good shield against harmful rays to avoid sunburn (one's skin turning a little pink is already a sign of damage).

Regularly checking your moles is another important action. A simple way to do so is to follow the ABCDE rule for melanoma detection, which stands for asymmetry, border, colour, diameter and elevation (or evolution). If in doubt about the appearance of a mole, consult your doctor.

The effects of the heat are also to be considered – the poster below summarises the best ways to minimise its impact on health.

More information about sunshine risks and prevention, as well as the ABCDE rule, can be found [here](#).

Medical Service

Exclusive CERN shop discounts

CERN access card holders and event participants can benefit from exclusive discounts all year long at the [CERN shop](#) in Science Gateway from Tuesday to Sunday, 8.30 a.m. to 5.30 p.m.

- **10% for CERN access card holders** (members of personnel, contractors, family members, club members etc.)
Present your CERN access card (plastic RFID card with barcode) at the checkout to benefit from a 10% discount
- **10% for participants of events** on the CERN site and at Science Gateway

Event organisers must contact shop@cern.ch at least 24 hours before their event, sharing a copy of the event badge.

Participants must show their event badge at the checkout to benefit from a 10% discount

- **Discounts for bulk purchases** (for example conference and event items)
Organisers must contact shop@cern.ch with details of the order to receive a discount.
- For bulk purchases of books or purchases of books that are not available in the CERN shop, CERN personnel can also contact the CERN [Bookshop](#) for a quote, check preferential

rates and place orders for books for professional activities. The [complete Bookshop](#)

[catalogue](#) is available online; other titles are available [on request](#).

Please note: discounts cannot be combined.

The IMPACT tool is evolving

Find out more about the progressive changes to this important web application used for activity coordination and associated safety aspects

[IMPACT](#) (Intervention Management Planning and Coordination Tool) is a critical web application used by a large and diverse population at CERN to manage the Organization's permit-to-work processes, covering both safety- and coordination-related aspects. It standardises the way activities are declared across all CERN installations and its use is, in many cases, essential in order to obtain access to specific locations.

First launched in 2011, IMPACT has since become an essential tool at CERN with some 15 000 requests created every year.

Today, as some of the technologies used in the current version have reached their end-of-life, the tool's software is being redeveloped, offering optimisations while maintaining the main functionalities that are in daily use. These crucial updates are being coordinated by the Intervention Management Steering Board, which comprises representatives from all departments and large experiments, as well as the FAP-BC and EN-IM groups. The Board is carrying out extensive consultation among all IMPACT users in order to determine their requirements.

What is changing?

While the features and the processes remain largely the same, the user interface is being completely overhauled, and the tool will be usable

on mobile devices, such as smartphones and tablets. All existing data will be migrated and visible on the new system.

As an IMPACT user, what should I be aware of?

Features are gradually being deployed: [Note de Coupure](#), [Fire Permit](#) and [IS37](#) are already effective and the Activity feature will follow soon, along with other safety tasks. The new features (e.g. Fire Permits) are no longer linked to EDH. The status of the approvals for these documents will only be visible on the new IMPACT interface. The [Task Dashboard](#) shows all current and past tasks (e.g. approvals).

Keep informed about upcoming changes by subscribing to the [dedicated notification channel](#), or contact the team via the [ServiceNow form](#) if you have any questions about the updated tool.

The Intervention Management Steering Board,
Eloy Reguero Fuentes (FAP-BC) and Giovanni
Chierico (EN-IM)*

** Intervention Management Steering Board members: Anne-Laure Perrot (Chair) (EN), Jesper Nielsen (BE), Jean-Philippe Tock (EN), David Widegren (EN-IM GL), Martin Jaekel (EP), Olivier Prouteau (HSE), John Shade (IT), Denis Pazem (SCE), Jan Borburgh (SY), Mirko Pojer (TE).*

Obituaries

Mats Lindroos (1961 – 2024)

Mats Lindroos, who made major contributions to accelerator technology, passed away on 2 May 2024 aged just 62. Mats received his PhD in

subatomic physics from Chalmers University of Technology in Gothenburg, Sweden, in 1993 under the supervision of Björn Jonson. As a PhD student

he studied decay properties and hyperfine interactions from oriented nuclei, making use of the low-temperature nuclear orientation facilities at ISOLDE, Daresbury and Studsvik. He joined CERN as a research fellow in 1993 and became a staff member in 1995.



(Image: Lindroos)

While at CERN, Mats performed a number of diverse roles, including being responsible for PS Booster operation and the technical coordination of the ISOLDE facility. He was one of the driving forces behind the HIE-ISOLDE project, which commenced construction in 2009 and is now one of the major accelerated-radioactive-beam facilities worldwide. While at CERN he also played leading roles in several European-Union-supported design studies for future conceptual accelerator facilities: the nuclear-physics radioactive-beam facility EURISOL and the beta-beam neutrino factory.

In 2009, when Sweden and Denmark were selected to be the host countries for the European Spallation Source (ESS), Mats returned to his roots in Sweden on secondment from CERN, formally

joining the ESS in 2015. As one of the earliest members of the ESS organisation, he was responsible for establishing the nascent accelerator organisation as well as the accelerator collaboration, set up as a CERN-like collaboration, between major European accelerator laboratories across 10 countries to undertake the technical design of this important part of the facility. Mats led the technical design for the 5-MW proton linac of the ESS, and from 2013, as head of the 100-strong accelerator division, he led the linac project that is now in the late stages of construction and installation. Even after stepping down from his leadership roles because of illness, he enthusiastically accepted a new one to advise the ESS management. He was fully involved in the process, and undoubtedly would have been instrumental in guiding the future evolution of the facility.

As a globally recognised expert on accelerator technology, Mats served on many committees in an advisory role, such as the IJCLab strategic advisory board (France), IN2P3 scientific committee (France), J-PARC technical advisory committee (Japan), the PIP-II Fermilab technical advisory committee (US) and CERN's Scientific Policy Committee. As an adjunct professor at Lund University he enjoyed teaching and supervising students in addition to his numerous research, management and committee roles.

Despite all these work activities, Mats found time to oversee, together with his partner Anette, the construction of a house on the south Swedish coast, where they enjoyed walking, gardening and being active in the local community.

Mats has touched all our lives with his energy and passion for research, his creativity for new ideas, his worldly knowledge, his sense of humour and, most importantly, his humanity and kindness. He will be greatly missed by all of us who had the privilege to count him as a friend and colleague.

His friends and colleagues

This obituary will also appear in the July/August CERN Courier.

Hans Joachim Specht (1936 – 2024)



(Image: Specht)

Hans Joachim Specht, one of the founders of ultra-relativistic heavy-ion physics and a pioneering figure in hadron cancer therapy, passed away on 20 May 2024, at the age of 87. A graduate of the University of Munich and ETH Zurich, and full Professor at the University of Heidelberg for more than 30 years, his career was distinguished by important contributions across a spectrum of scientific domains.

Hans started his academic career in atomic and nuclear physics in Munich, under the guidance of Heinz Maier-Leibnitz. A highlight was the discovery and precise measurement of shape isomerism in heavy nuclei. His observation of distinct rotational bands in ^{240}Pu showed, for the first time, that nuclei can be in a strongly deformed cigar-shaped state shortly before fission, confirming the concept of a “double-humped” fission barrier. In Munich, and later in Heidelberg, he developed several innovative large-scale detectors for fission fragments and reaction products of heavy-ion collisions, becoming one of the leading experimentalists in the new field of “heavy-ion physics”, with experiments at the MPI for Nuclear

Physics in Heidelberg and at the newly founded GSI in Darmstadt.

In the early 1980s, Hans reoriented his research towards the higher energies available at CERN. His contributions and advocacy, alongside a handful of other enthusiastic proponents, were instrumental in establishing CERN’s ultra-relativistic heavy-ion programme at the SPS accelerator, which was approved in 1984. He became the spokesperson of a first-generation heavy-ion experiment (Helios/NA34-2), initiator and spokesperson of a second-generation experiment (CERES/NA45), and a crucial supporter and mentor for the third-generation ALICE experiment at the Large Hadron Collider.

Initially with the bespoke dilepton experiment CERES, and later as a leading force within NA60, Hans succeeded in detecting, for the first time, thermally produced lepton pairs in heavy-ion collisions. Arguably one of the most challenging signals, the “Planck-like” spectrum of thermal radiation at higher masses, and the precise characterisation of the in-medium modification of the rho-meson at lower masses, proved to be crucial in establishing the existence and properties of the quark–gluon plasma – the state of strongly interacting matter thought to have existed in the primordial Universe, just a few microseconds after the Big Bang. The enduring quality and relevance of these measurements remain unsurpassed almost two decades later.

Hans also provided forward-looking impulses for nuclear and particle physics research as Scientific Director of GSI (1992–1999), where he set the technical and science-policy course for the development and application of a groundbreaking innovation in radiation medicine – ion-beam cancer therapy. A pilot project at GSI for the irradiation of tumours with ^{12}C ions, launched under his leadership, successfully treated 450 patients and led to the establishment of the Heidelberg Ion-Beam Therapy Center (HIT), the first European ion-beam therapy facility. Reflecting on his achievements, he was most proud of his contributions to ion-beam therapy, improving human health through scientific advancement.

Hans also had a profound interest in the intersection of physics, music and neuroscience, collaborating with Hans-Günter Dosch on understanding perception of music and its physiological bases. This transdisciplinary approach produced highly cited publications on the differences in the auditory cortex between musicians and non-musicians, expanding the boundaries of how we understand the brain and its response to music.

Hans was an outstanding teacher, a prolific mentor, a successful science manager, but

foremost, he was an inquisitive scientist – someone who profoundly loved physics, with a thirst for knowledge, a hunger for understanding and a relentless drive to follow wherever his interests and research would lead him.

His friends and colleagues

A full obituary will appear later in the year in the CERN Courier.

Ombud's Corner

First steps in the world of work

As the new CERN Ombud, I attended the [information session](#) for all new employed and associated members of the CERN personnel held on 11 June. There, I had the opportunity to meet students who were taking their first steps in the world of work.

They were enthusiastic and curious, and they expressed many expectations, some of them high. It's true that a lot is at stake for them. An experience like this can influence their studies and even the course of their lives. It reminds me of my first internship, which made me realise that translation wasn't for me and led me to change my career path.

Students with little or no professional experience may be taken aback when they find themselves in unfamiliar situations. This was the case for one student who recently knocked on my door: *"I don't have any professional experience, so I was left speechless when my supervisor criticised how I organise my work, even though I'm doing a good job"*.

Thanks to our conversation, the student was able to take stock and realise that she needed to be given more autonomy and more flexibility in the way she organised herself on a daily basis in order to perform to the best of her ability. Following this period of reflection, she was able to discuss her expectations with her supervisor and to calmly accept what was expected of her.

Our working relationships require regular adjustments throughout our professional lives. As the above example shows, we must first take stock of our own expectations, needs and possible fears in order to decide how we would like to interact with others, and how we want others to interact with us. This also makes others more likely to respect us and appreciate our collaboration. Regardless of its impact, what matters most is that this first experience of the working world goes smoothly and that the students, with the support of their colleagues, feel that they have progressed and made a valuable contribution.

Marie-Luce Falipou

I would like to hear your reactions and suggestions – join the CERN Ombud Mattermost channel at: [Town Square - CERN Ombud CERN Mattermost](#)
The Ombud is available from Monday to Friday in office B500/1-004 on the Meyrin site. To make an appointment, in person or online, contact the Ombud at ombuds@cern.ch.

More information can be found on the Ombud's website: <https://ombuds.web.cern.ch/>

The Staff Association is launching a survey on well-being at work. You can take part by completing the [online questionnaire](#), which is available up until and including July 3.